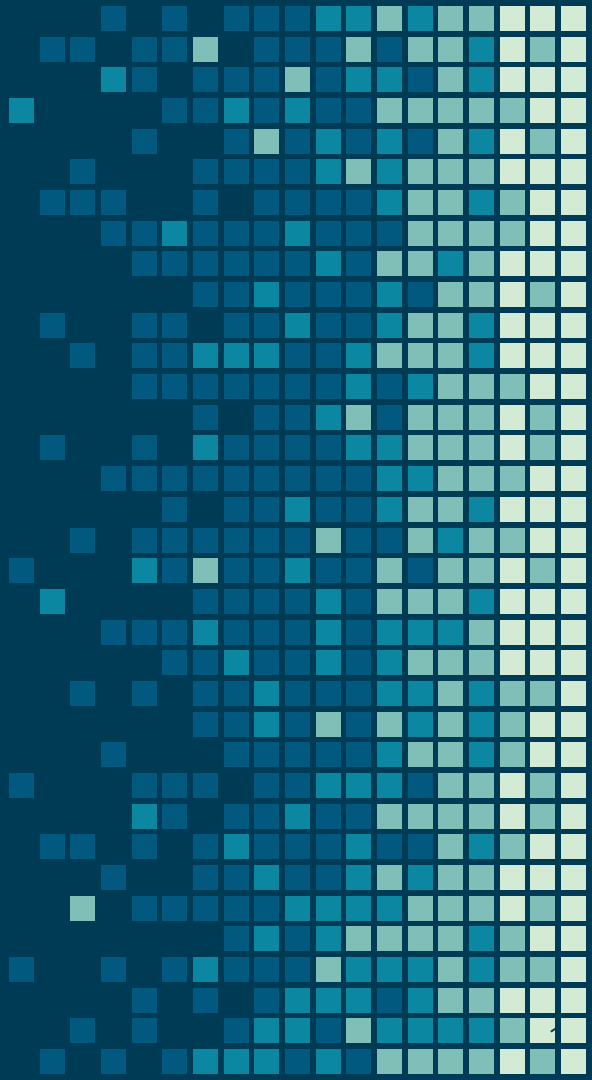
EMBEDDED CONTROL

Prof. Luca De Cicco
luca.decicco@poliba.it
https://c3lab.poliba.it/LDC



Debugging with SWD

Debugging through SWD/SWV

- Serial Wire Debug (SWD) is a two-pin debug interface used to control the MCU during debugging (run, stop on breakpoints, single-step) .
- Serial Wire Output (SWO) is an optional trace output pin used to send real-time trace data from the MCU to the debugger.
- Serial Wire Viewer (SWV) is the trace technology that uses SWD and SWO to display real-time information such as printf() output, exceptions, program counter samples, and variable changes.

Why printf() through UART is inefficient

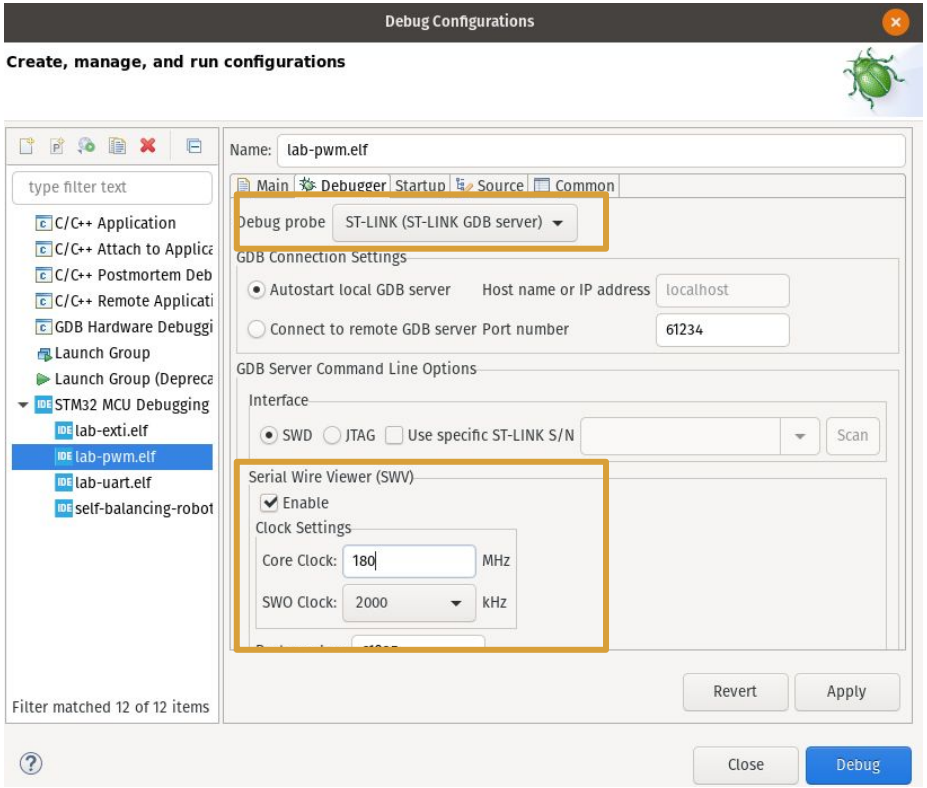
- `printf()` statements are a common, but inefficient, way to debug firmware
- A typical approach is to send `printf()` output from the MCU to the host PC through a UART interface
- This is inefficient since transmitting each character through UART can require many CPU cycles and interfere with the real-time behavior of the application
- Arm Cortex-M processors provide a very low-overhead mechanism to send data from the processor core to the debugger through the SWO pin
- This mechanism uses the Instrumentation Trace Macrocell (ITM), which writes trace data to memory-mapped stimulus ports

Writing to the debugger through ITM

- ITM can be used to redirect printf() output to a console view in the debugger (STM32CubeIDE)
- The overhead is low because data is written to memory-mapped ITM stimulus ports
- The ITM provides 32 stimulus ports, also called channels.
- By writing different types of data to different ITM stimulus ports, the debugger can display or process each data stream differently.

Enable SWV debugging

- Open debug configuration
- Select “ST-LINK” as “Debug probe”
- Enable “Serial Wire Viewer”
- Set the Core Clock to match the core clock frequency in MHz, the click apply



Make your firmware SWD/ITM ready

To redirect `printf()` output to ITM stimulus port 0, redefine the `__io_putchar()` weak function:

```
#include <stdlib.h>
#include <stdio.h>
#include "stm32f4xx.h"
/* _write uses ITM_SendChar to write through the SWD interface */
int __io_putchar(int ch)
{
   ITM_SendChar(ch);
    return ch;
}
```

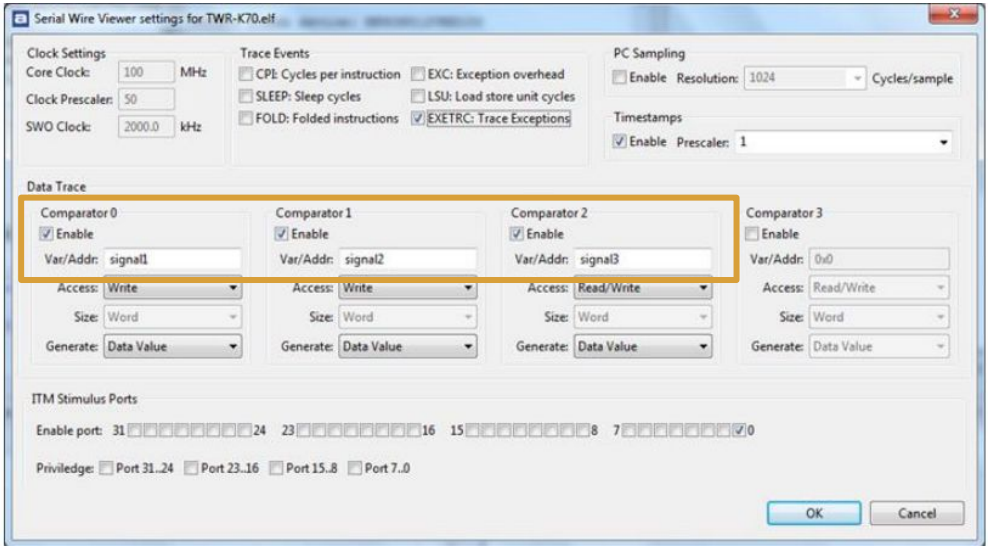
Where `printf()` output is shown?

- The SWV ITM Data console shows the output of printf()
- When running the firmware in debug mode first click on the "record" red button
- Then configure the ITM trace by clicking on the wrench icon
- Select channel 0 in "ITM stimulus port"



Inspecting variables through SWV

Global variables can be continuously and graphically inspected by inserting the var name in the “Comparator X” var/addr field



Example of time evolution of a variable

The SWV trace timeline graph shows the content of a variable (or more than one variable) automatically and continuously

